higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

5 December 2024

This marking guideline consists of 8 pages.

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SECTION A

QUESTION 1

1.1 B✓✓
1.2 A✓✓
1.3 C✓✓
1.4 A✓✓
1.5 C✓✓

(5 × 2) **[10]**

QUESTION 2

2.1 True
2.2 False
2.3 False
2.4 True
2.5 True
2.6 True
2.7 False
2.8 False

(8 × 1) **[8]**

QUESTION 3

3.1 B
3.2 E
3.3 C
3.4 A
3.5 F
3.6 G
3.7 D

(7 × 1) **[7]**

TOTAL SECTION A: 25

SECTION B

QUESTION 4

4.1

						BINARY							
POS			1	1	1	0	1,	0	1	1	1	1	
NEG				1	1	1	0,	1	0	0	1		
						BINARY							
POS		0*	0	1	1	1	0	1,	0	1	1	1	1	
NEG		1*	0	0	1	1	1	0,	1	0	0	1	0	
							1'S						
POS		0*	0	1	1	1	0	1,	0	1	1	1	1	✓
NEG		1*	1	1	0	0	0	1,	0	1	1	0	1	✓
	1	0*	0	0	1	1	1	0	1	1	1	0	0	✓
													1	
		0*	0	0	1	1	1	0,	1	1	1	0	1	✓
PLUS					1	1	1	0,	1	1	1	0	1	✓

(5)

4.2

					BINARY							
POS			1	0	1,	0	1	1				
NEG		1	1	0	1	0,	1	0	0	1		
					BINARY							
	0*	0	0	1	0	1,	0	1	1	0		
	1*	1	1	0	1	0,	1	0	0	1		
						2'S						
	0*	0	0	1	0	1,	0	1	1	0	✓	
	1*	0	0	1	0	1,	0	1	1	1	✓	
0	1*	0	1	0	1	0,	1	1	0	1	✓	
		1	0	1	0	1,	0	0	1	0		
PLUS												
1										1	✓	
NEG		1	0	1	0	1,	0	0	1	1	✓	

(5)

4.3

						1	0	0	0	1	1,	1	1	1	
MULTIPLY									1	1	0,	1	0	1	
						1	0	0	0	1	1	1	1	1	✓
				1	0	0	0	1	1	1	1	1			✓
			1	0	0	0	1	1	1	1	1				✓
		1	0	0	0	1	1	1	1	1					✓
EQUAL		1	1	1	0	1	1	0	1,	1	0	1	0	1	1	✓

(5)

4.4

										1	1	1	1	0,	1	1	0	1
1	0	1,	0	1	1	1	0	1	0	0	1	0	1,	1	0	0	1	1
						1	0	1	0	1	1							
						1	0	0	1	1	1	1						
							1	0	1	0	1	1						
						1	0	0	1	0	0	1						
							1	0	1	0	1	1						
								1	1	1	1	0	0					
								1	0	1	0	1	1					
									1	0	0	0	1	0	1			
										1	0	1	0	1	1			
											1	1	0	1	0	1		
											1	0	1	0	1	1		
													1	0	1	0	1	1
													1	0	1	0	1	1
																		0

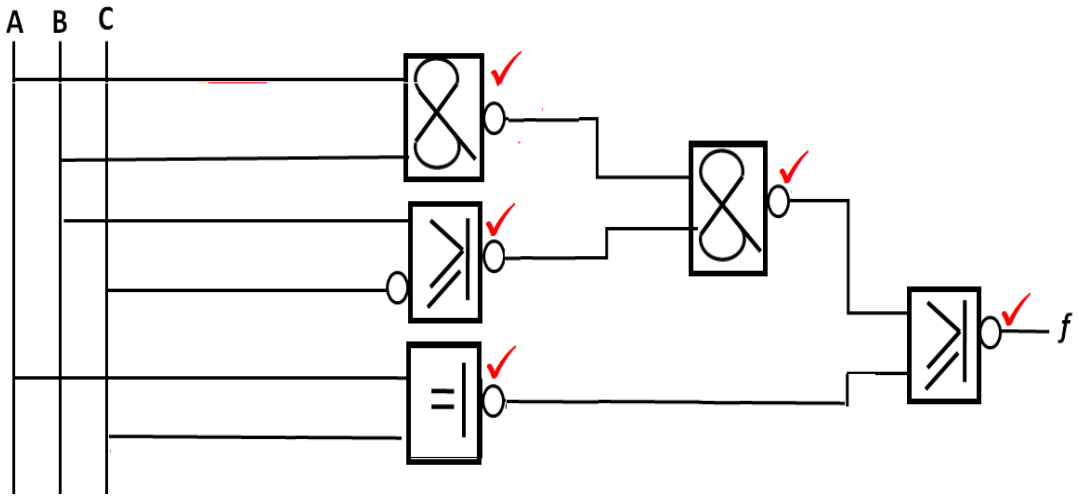
✓✓ ✓ ✓ ✓

(5)

[20]

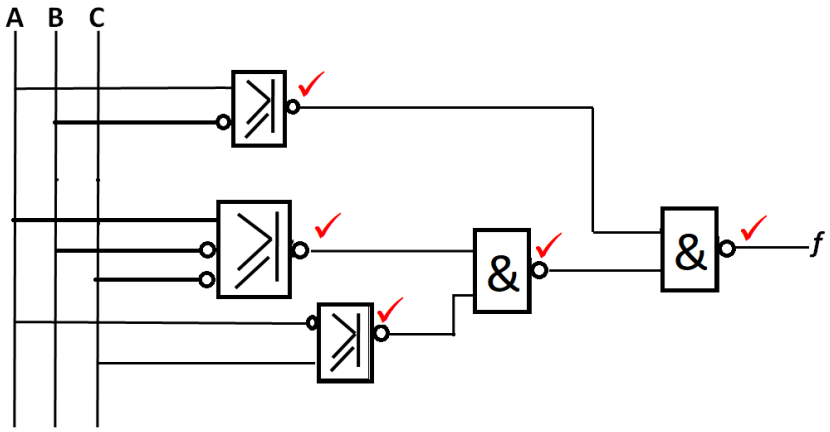
QUESTION 5

5.1



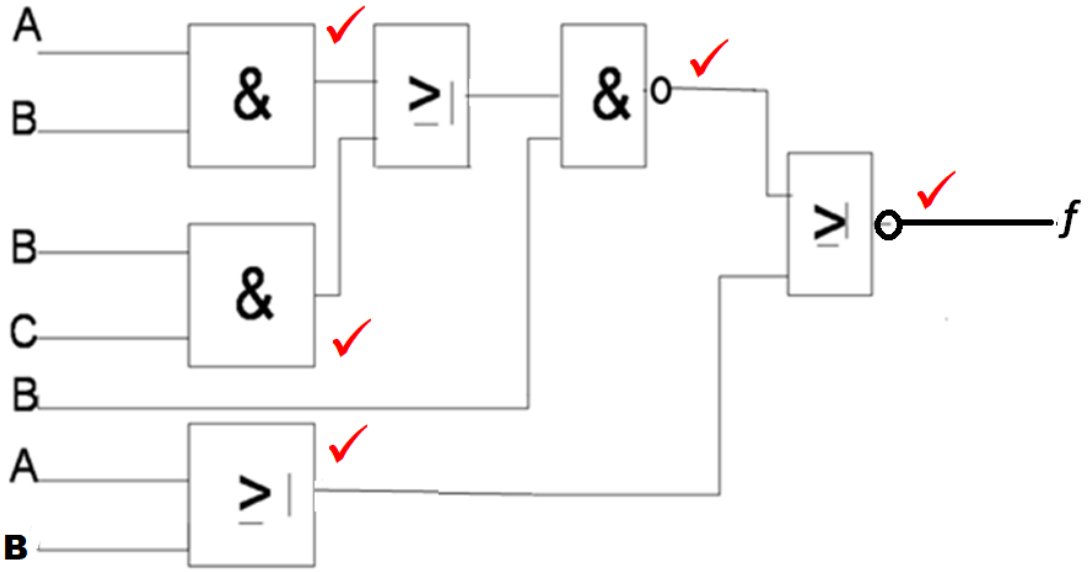
(5)

5.2



(5)

5.3



(5)
[15]

QUESTION 6

A	B	C	*f*
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

✓

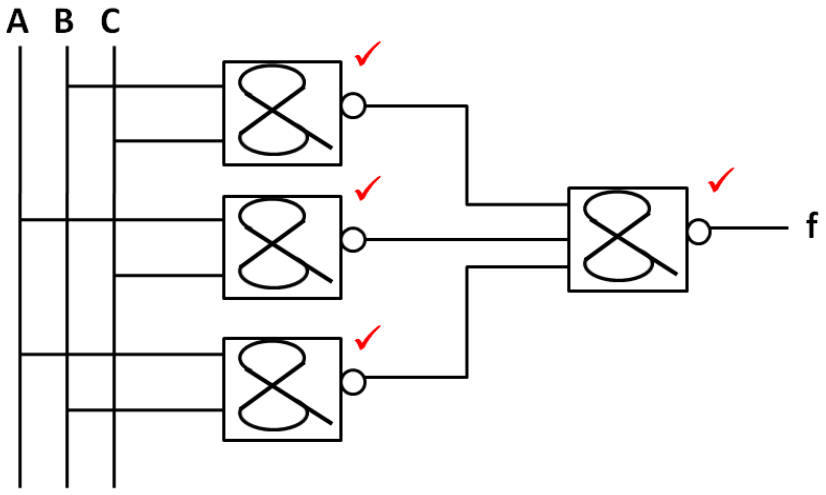
$$\overline{A}BC+A\overline{B}C+AB\overline{C}+ABC=f\checkmark$$

$$\overline{A}BC+A\overline{B}C+AB=f\checkmark$$

$$\overline{A}BC+AC+AB=f\checkmark$$

$$BC+AC+AB=f\checkmark$$

$$\overline{\overline{BC}\,.\,\overline{AC}\,.\,\overline{AB}}=f\checkmark$$



✓ ✓ ✓ ✓

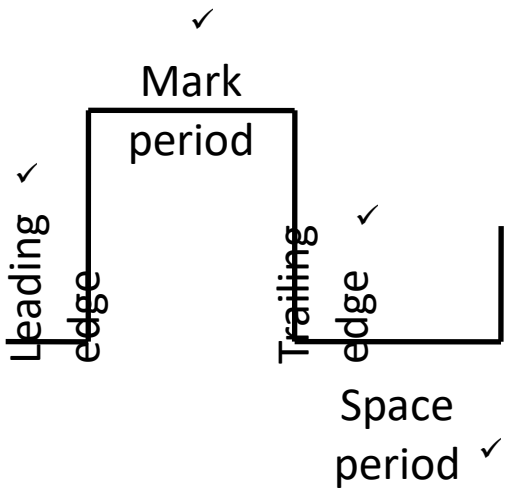
[10]

QUESTION 7

7.1 Fan-out is a term that defines the maximum✓ number of digital inputs✓ that the output✓ of a single logic gate can feed.✓ (4)

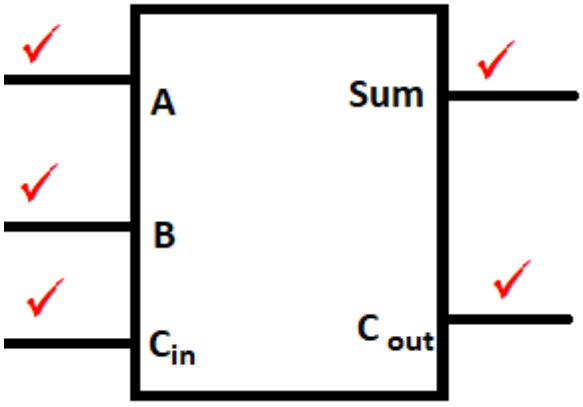
7.2 8 (1)

7.3



(4)

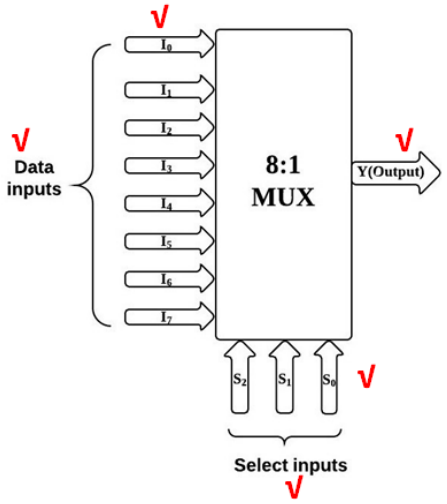
7.4



A	B	C/I	SUM	C/O	
0	0	0	0	0	✓
0	0	1	1	0	
0	1	0	1	0	✓
0	1	1	0	1	
1	0	0	1	0	✓
1	0	1	0	1	
1	1	0	0	1	✓
1	1	1	1	1	

(9)

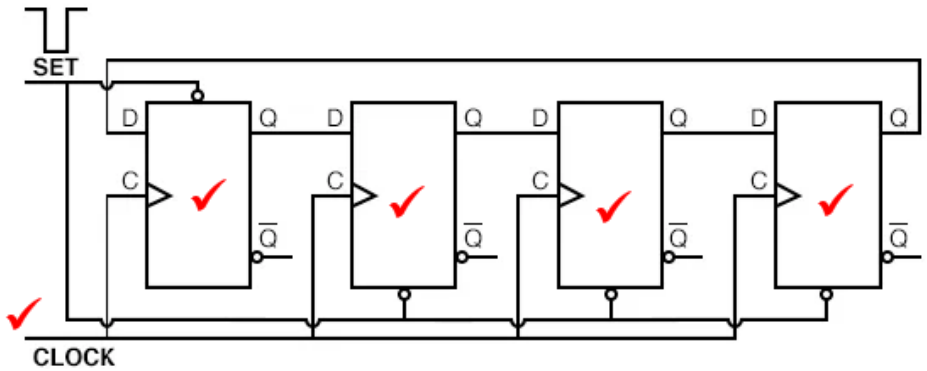
7.5



Allocate 1 mark for the input lines
Allocate 1 mark for the output lines
Allocate 1 mark for the select lines
Allocate 1 mark if 8 input lines are indicated
Allocate 1 mark if 3 select lines are indicated

The purpose of the select lines is to specify which one✓ of the input lines are connected✓ to the output line. (7)

7.6



(5)
[30]

TOTAL SECTION B: 75
GRAND TOTAL: 100

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